

One²: An Intrusion Detection System for Both Internal and External Vehicular Network From Weak Labeled Data

He Lu , Yilong Ren , Yue Liu, Yanan Zhao , Kaiwen Guo , Yang Yang, and Haohua Du 

Abstract—Network attacks on the Internet of Vehicles (IoV) can lead to catastrophic consequences such as traffic congestion, incorrect routing, and even accidents. Existing rule-based countermeasures are effective only in specific scenarios, while machine learning-based methods suffer from suboptimal performance due to data quality issues. Furthermore, the isolation between in-vehicle networks (IVN) and external vehicle networks (EVN) prevents independent intrusion detection systems (IDS) from detecting continuous attack behaviors. To address these issues, we propose a mechanism that characterizes attack behaviors solely based on network connection information. This mechanism downplays the specifics of the traffic itself, allowing for the integration of IVN and EVN through structure, thus breaking the internal-external boundary and providing relatively stable logical structure information. This approach also offers opportunities to address the scarcity of labeled data. Based on this characterization mechanism, we develop an intrusion detection system – One², utilizing a multi-attribute heterogeneous graph transformer to achieve accurate multi-classification of various types of attacks. To assess the compatibility of the proposed IDS for IVN and EVN, extensive experiments were conducted using six real-world datasets that accurately depict IVN and EVN. The results show that One² improves average accuracy by 7.14% and F1 score by 6.97% compared to state-of-the-art methods.

Index Terms—Internet of Vehicles (IoV), intrusion detection system (IDS), weak label.

Received 9 April 2025; revised 22 September 2025; accepted 3 November 2025. Date of publication 6 November 2025; date of current version 12 March 2026. This work was supported in part by the National Key R&D Program of China under Grant 2022YFB3104403 and in part by Shandong Provincial Natural Science Foundation under Grant ZR2025QB43 and Grant ZR2025QC1536. (Corresponding author: Haohua Du.)

He Lu is with the School of Civil and Environmental Engineering, Nanyang Technological University, Singapore 639798, and also with the School of Transportation Science and Engineering, Beihang University, Beijing 100191, China (e-mail: herlu@buaa.edu.cn).

Yilong Ren is with the School of Transportation Science and Engineering, Beihang University, Beijing 100191, China, also with the Beihang Hangzhou Innovation Institute, Hangzhou 310023, China, also with Zhongguancun Laboratory, Beijing 100083, China, and also with the State Key Lab of Intelligent Transportation System, Beijing 100191, China (e-mail: yilongren@buaa.edu.cn).

Yue Liu is with the School of Urban Construction, Beijing University of Technology, Beijing 100124, China (e-mail: yueliu@emails.bjut.edu.cn).

Yanan Zhao and Yang Yang are with the School of Transportation Science and Engineering, Beihang University, Beijing 100191, China (e-mail: zynbyxz@gmail.com; yang_yang@buaa.edu.cn).

Kaiwen Guo is with the Ocean University of China, Qingdao 266100, China (e-mail: kevinguo@ouc.edu.cn).

Haohua Du is with the School of Cyber Science and Technology, Beihang University, Beijing 100191, China (e-mail: duhaohua@buaa.edu.cn).

The source code of this work is available at: <https://github.com/LouHerGetUp/One2>

Digital Object Identifier 10.1109/TDSC.2025.3629862

I. INTRODUCTION

THE rapid development of the Internet of Vehicles (IoV) enables vehicles to communicate with various entities, including network infrastructure and pedestrians [1]. While this connectivity brings convenience, it also introduces significant security risks [2]. For example, renowned white hat hackers Valasek and Miller exploited a vulnerability in the in-vehicle infotainment (IVI) system to gain remote control of a Jeep Cherokee [3]. Similarly, Keen Security Lab of Tencent demonstrated a comprehensive attack chain that successfully infiltrated the in-vehicle network (IVN) system through the wireless communication module of the external vehicle network (EVN), enabling remote control attacks on both Tesla Model series and BMW vehicles [4], [5]. If such incidents were to occur in a real traffic environment, they would significantly compromise the vehicle's communication safety and functional integrity, even leading to traffic accidents and casualties. Therefore, Intrusion Detection Systems (IDS) for IoV have become a critical research issue recently [6].

Currently, intrusion detection systems (IDS) for the Internet of Vehicles (IoV) can be categorized into two main types: rule-based and machine learning-based. Rule-based IDS can only describe known and formalizable attack behaviors, making them effective only in specific scenarios [7], [8], [9]. In contrast, machine learning-based IDS automatically derive “rules” from data, allowing them to be applicable in more diverse scenarios [10], [11], [12]. However, the performance of machine learning algorithms is highly dependent on data quality, which limits the effectiveness of these systems [13], [14]. Recent efforts have utilized self-supervised learning to address the issue of scarce labeled data in IoV scenarios [15]. However, these efforts typically focus only on either the IVN or the EVN. Given that attack chains typically progress from EVN to IVN, detecting intrusions at each network layer separately is crucial. Early detection of attacks at the EVN level enables preemptive warnings and protective measures, thereby preventing attackers from progressing to the IVN and gaining control of the vehicle. Therefore, it is evident that an IDS system for IoV needs to have three key capabilities: **recognizing attack types**, being **compatible with both IVN and EVN**, and **performing well with limited labeled data**. However, achieving these capabilities is highly challenging due to the following reasons:

- 1) Attack behaviors are often multilayered and hidden within normal behaviors, leading to similar features among different attack types and diverse features within the same attack type. This makes feature-based recognition extremely difficult.
- 2) IVN and EVN use different network protocols, resulting in distinct network data structures, complicating the design of an intrusion detection algorithm that works for both.
- 3) Machine learning-based methods heavily rely on data quality for performance. IoV data is difficult to obtain due to its sensitive nature, and the few available datasets often lack high-quality type annotations.

To address the aforementioned challenges, we propose a behavior representation mechanism based on network structure, modeling network data as a multi-attribute heterogeneous graph. In this graph, data flows, IP addresses, and ports are treated as nodes, and nodes belonging to the same traffic flow form a fully connected subgraph. This approach weakens the specific characteristics of the traffic data, allowing the logical integration of IVN and EVN by breaking the boundaries between them. Additionally, the stability of the structure serves as implicit information, enabling the clustering of behavior data with similar structures for self-supervised learning with weak labels [16]. Furthermore, to effectively learn the information from this multi-attribute heterogeneous graph, we introduce a corresponding Multi-attribute Heterogeneous Graph Transformer (MA-HGT). This model employs multiple attention heads to learn different attention weight distributions within the graph and capture correlations at various positions [17], thereby enhancing the model's representational capability.

In summary, the major contributions of this work are as follows:

- We designed and implemented an IDS called One² that is compatible with both IVN and EVN using limited weakly labeled data. It can accurately identify attacks, including cross-network continuous attacks.
- We proposed a network structure-based representation mechanism, modeling network data as a multi-attribute heterogeneous graph to extract deep implicit information. Additionally, we designed a novel deep learning model structure, MA-HGT, to efficiently learn the information within this multi-attribute heterogeneous graph.
- To validate the performance of One², we conducted experiments on multiple real-world datasets. The results show that One² improves average accuracy by 7.14% and F1-Score by 6.97% compared to state-of-the-art methods.

II. RELATED WORKS

A. Machine Learning-Based IDS

Machine learning [18] algorithms have been widely used in IDS. Alshammari et al. [19] developed an IDS for IVN using the two algorithms. Derhab et al. [20] proposed a Histogram-based Intrusion Detection. However, these IDS exhibit high False Alarm Rates (FAR) and are exclusively suitable for IVN. To safeguard IVN and EVN and reduce the FAR, Yang et al. [14] integrated four distinct ML methods to develop a novel IDS.

In [21], the authors also designed an IDS incorporating three algorithms. Nevertheless, both IDS heavily rely on feature engineering. Additionally, they require extensive training using a substantial volume of labeled data.

B. Deep Learning-Based IDS

To avoid the dependence on feature engineering, researchers employ Deep Learning [22] to build IDS. Seo et al. [23] proposed a Generative Adversarial Network-based IDS for IVN. Nguyen et al. [24] proposed a novel approach for developing an IDS specifically designed for CAN buses. However, these IDS fail to consider the intricate attributes of the external network. As a result, their effectiveness is limited to IVN with relatively simple data structures. For EVN, Han et al. [25] introduced the Spatial and Time-aware Intrusion Detection Model. Deng et al. [15] proposed an IDS based on Graph Convolutional Network with stream topology. To achieve Bot detection in external networks, Zhao et al. [26] transformed the bot detection problem into a classification task on a semi-supervised graph. To detect Distributed Denial of Service (DDoS) attacks, Cao et al. [27] designed an IDS based on a spatio-temporal graph convolutional network. However, these IDS only apply to EVN and mostly focus on the binary classification problem of network traffic. To design IDS that can be used in IVN and EVN, Yang and Shami [28] proposed an IDS system based on transfer learning and ensemble learning. However, it is important to note that transforming network data into images for training requires substantial data. Ashraf et al. [13] developed an IDS utilizing the LSTM autoencoder architecture. Nevertheless, these two IDS only address the binary classification problem of IoV traffic data.

III. IOV AND THEIR VULNERABILITIES

A. Preliminary

The IoV enables wireless communication between vehicles and network infrastructure setups, making it a critical foundation for intelligent transportation systems. According to the communication range, the IoV can be categorized into IVN and EVN [29]. The IVN consists of numerous Electronic Control Unit (ECU) interconnected via controller area network (CAN) bus [30]. This facilitates the integration of various features in Connected Vehicles (CV) and Autonomous Vehicles (AV) [31]. The EVN establishes a connection between CV, AV and external devices via Vehicle-to-Everything (V2X) technology, including Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P) and Vehicle-to-Sensors (V2S) [2].

B. Vulnerabilities of IVN

In IVN, ECU play a crucial role in facilitating the execution of diverse functionalities by CV and AV. Consequently, an increasing number of ECU are being integrated into contemporary vehicles, with many luxury vehicles boasting an excess of 100 ECU [30]. Typically, These ECU establish communication through the CAN bus. Fig. 1 illustrates the structure of a CAN packet, which consists of seven distinct fields [32]. Among these fields, the data field holds the information that

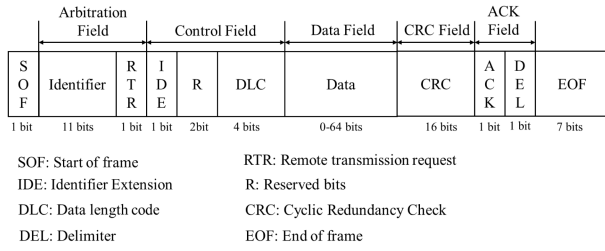


Fig. 1. CAN packet structure [13].

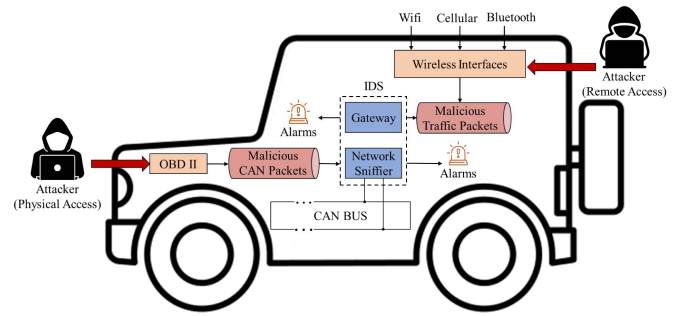
governs the functioning of the node, making it the most crucial and susceptible component [14]. Message injection attack is a prominent form of attack in IVN, which is realized by an attacker intentionally inserting malicious messages into the data field of CAN packets. This manipulation is aimed at gaining control over the vehicle. Message injection attacks can be classified into three main categories: Denial of Service (DoS), fuzzy, and spoofing attacks [33]. DoS attacks flood the CAN with a substantial volume of messages, leading to the delay or unavailability of valid messages. In a similar vein, fuzzy attacks induce the vehicle to have malfunctions or display unanticipated behavior by the injection of random packets. Spoofing attacks involve the deliberate injection of targeted packets with the aim of disguising them as legitimate signals, hence enabling unauthorized control over the vehicle.

C. Vulnerabilities of EVN

With the increasing number of entities accessing IoV, EVN are becoming enormous networks containing diverse networks and devices [34]. Due to the lack of encryption and authentication mechanisms, EVN are vulnerable to a variety of traditional cyber threats, wherein every entity within the network possesses the potential to be targeted as an attack node. The major types of attacks in EVN are DoS, infiltration, sniffing, brute force, Web attacks, and botnet attacks [14], [35]. Among these, DoS attacks overwhelm the vehicle with a large number of messages, resulting in the vehicle's inability to handle legitimate requests. Infiltration attacks [35] refer to aiding other assaults by traversing a compromised protective system. Sniffing attacks collect sensitive information by capturing packets in the EVN. Brute force attacks are used to obtain malicious vehicle control by breaking passwords. Web attacks [35] are employed with the intention of executing detrimental actions by targeting the web interfaces of entities within the EVN. Botnets infect many vehicles and equipment with Bot viruses to facilitate assaults such as DoS.

D. Security Model of IVN and EVN

Fig. 2 illustrates the general attack scenarios against the CV or AV, encompassing both physical and remote attacks. A physical attack pertains to the act of an assailant utilizing computing equipment, such as a laptop, to get unauthorized access to the CAN bus by establishing a connection with the OBD-II port in IVN. This unauthorized access enables direct control over critical vehicle functions, including steering, braking, and acceleration systems. Remote attacks involve the exploitation of wireless interfaces in the EVN—such as WiFi, cellular networks, and

Fig. 2. Attack scenarios and proposed IDS – One² deployment.

Bluetooth—which attackers leverage to subsequently compromise IVN systems, creating cross-network attacks as discussed in Section III-C. The objective of these attacks is to compromise the telematics equipment and CAN bus of the vehicle [14]. Execution scenarios for this type of attack include V2V attacks and V2I vulnerabilities. (1) V2V attacks: Malicious vehicles can broadcast falsified safety messages (e.g., fake emergency braking warnings) or spoofed position information, messages, potentially causing traffic disruption or collisions. (2) V2I vulnerability: Attackers can compromise existing roadside devices or deploy malware on them. Through these compromised or spoofed infrastructure devices, an attacker can launch an attack directly at passing vehicles. This interconnected threat landscape demonstrates why unified security approaches spanning both IVN and EVN domains are essential for comprehensive vehicle protection. Attackers can exploit the boundaries between these networks, using compromised telematics equipment as a bridge to access critical CAN bus functions, or leveraging captured CAN data to orchestrate more sophisticated external attacks. Additionally, The declining costs of memory and computing devices have recently facilitated the widespread adoption of deep-learning-based IDS in CV and AV [13].

The proposed IDS deployment is shown in Fig. 2 To ensure the security of the IVN, the proposed IDS will be implemented directly on the CAN bus. The network sniffer intercepts and scrutinizes all flow data transmitted on the CAN bus to monitor and identify suspicious network activities. This capability is crucial for defending against physical access attacks [13], [36]. The proposed IDS will also be deployed within the gateway to safeguard the vehicle against remote attacks initiated from the EVN. The IDS at the gateway will detect all flow data transmitted to and from the CAN bus originating from the EVN, thereby establishing a defense against remote access attacks [14]. In the event that an attacker sending malicious traffic to the car is detected, the proposed IDS will promptly raise the alarm and prevent the attacker from gaining access, thereby protecting the vehicle from potential damage caused by cyber-attacks.

IV. THE IDS: ONE²

A. System Architecture

The One² framework has three steps, as shown in Fig. 3. First, network flow data is constructed into the graph construction. Subsequently, the graph construction is employed as the input

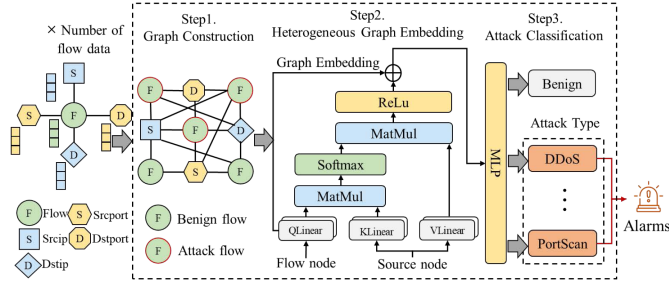


Fig. 3. Framework of the proposed IDS.

 TABLE I
SOME ESSENTIAL ATTRIBUTES OF IVN FLOW DATA

Name	Type	Description
ID	Integer	Identifier of internal network flow data
Time	Timestamp	Internal network flow start time
DLC	Integer	Number of data bytes
DATA	Integer	Data value of network flow data
Label	Integer	Label of benign data or attack types

for MA-HGT to perform graph embedding. Ultimately, the flow nodes are classified based on the graph embedding. Additionally, to enhance training efficiency when processing large-scale intrusion detection data, an effective heterogeneous graph sampling algorithm is employed to sample the graph structure during training.

B. Graph Construction

Definition 1. Heterogeneous Graph: A heterogeneous graph is defined as a graph with multiple types of nodes and/or edges. We denote a heterogeneous graph as $G = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R})$, where $v \in \mathcal{V}$ denotes nodes, $e \in \mathcal{E}$ denotes edges, $\mathcal{A}$ denotes the union of node type, and $\mathcal{R}$ denotes the union of edge type.

Definition 2. Mapping function: The mapping function of node type and edge type is $\tau(v) : \mathcal{V} \rightarrow \mathcal{A}$ and $\phi(e) : \mathcal{E} \rightarrow \mathcal{R}$, respectively.

In real life, network flow data contain many attributes that may indicate potentially malicious activity. Hence, these attributes can be used to differentiate malicious attacks from benign behaviors. Tables I and II provide some essential attributes of IVN and EVN flow data, respectively. Instead of traditional IDS that directly input attributes into the model, the proposed IDS adopts a novel approach by constructing a graph representation of network data before the model learning process. In detail, we build the One² of network flow data for the first time. This graph enables us to describe the network flow more fine-grained and capture implicit topological features of flow which facilitates accurate identification of malicious behavior in the network.

The reasons for using graph structures to build an IDS for IoV are as follows: (1) Unified representation of heterogeneous data: By modeling network data as a multi-attribute heterogeneous graph, data streams, IP addresses, and ports are uniformly represented as nodes. This allows IVN and EVN to be logically unified and processed together, breaking down the boundaries between

 TABLE II
SOME ESSENTIAL ATTRIBUTES OF EVN FLOW DATA

Name	Type	Description
Srcip	String	Source IP address
Dstip	String	Destination IP address
Srport	Integer	Source port number
Dstport	Integer	Destination port number
Time	Timestamp	EVN flow start time
Dur	Float	EVN flow total duration
Proto	Integer	transaction protocol
FwdPac	Integer	Total Forward Packets
BwdPac	Integer	Total Backward Packets
FwdPacLen	Integer	Total Length of Forward Packets
BwdPacLen	Integer	Total Length of Backward Packets
FwdHeLen	Integer	Forward Header Length
BwdHeLen	Integer	Backward Header Length
Label	Integer	Label of benign or attack types

them. (2) Adaptability to weakly labeled data: The inherent stability of graph structures provides implicit information, enabling the clustering of behavioral data exhibiting similar structural patterns, as shown in the left half of Fig. 4. This supports self-supervised learning approaches with weakly labeled data, thereby reducing dependence on extensive labeled datasets. (3) Effective capture of complex relationships and positional information: Nodes within the same traffic flow form fully connected subgraphs, enabling comprehensive modeling of inter-node relationships. Through the multi-attention head mechanism of the MA-HGT model, it learns positional associations within the graph structure. This enhances the model's representational capabilities, enabling it to capture complex patterns that traditional methods struggle to identify. (4) Preservation of structural features: In contrast to flattening or serialization approaches, the graph structure preserves the natural topological features of IoV network traffic. This structured representation maintains greater fidelity to the inherent characteristics of network data.

Given a IVN flow data I , we construct a multi-network attributes heterogeneous graph G_I with two types of nodes (*Flow* and *ID*) and one type of edges (*Flow* – *ID* edge). The identifier of IVN flow data is represented as a *ID* node, and the rest of the attributes of flow are represented as a *Flow* node. Moreover, we build an undirected edge which is from the *Flow* to the *ID*.

Given a EVN flow data E , we construct a multi-network attributes heterogeneous graph G_E with five types of nodes (*Flow*, *Srcip*, *Dstip*, *Srport* and *Dstport*) and four types of edges (*Flow* – *Srcip* edge, *Flow* – *Dstip* edge, *Flow* – *Srport* edge, *Flow* – *Dstport* edge). As illustrated in Fig. 3, the source IP address, destination IP address, source port number and destination port number of EVN flow data are denoted as a *Srcip* node, *Dstip* node, *Srport* node, *Dstport* node, respectively. Besides, the rest of the attributes of flow are denoted as a *Flow* node. After that, we build four undirected edges which are from the *Flow* to the *Srcip*, from the *Flow* to the *Dstip*, from the *Flow* to the *Srport*, from the *Flow* to the *Dstport*.

For the aforementioned graph construction methods, every *Flow* node in the graph represents a single piece of network

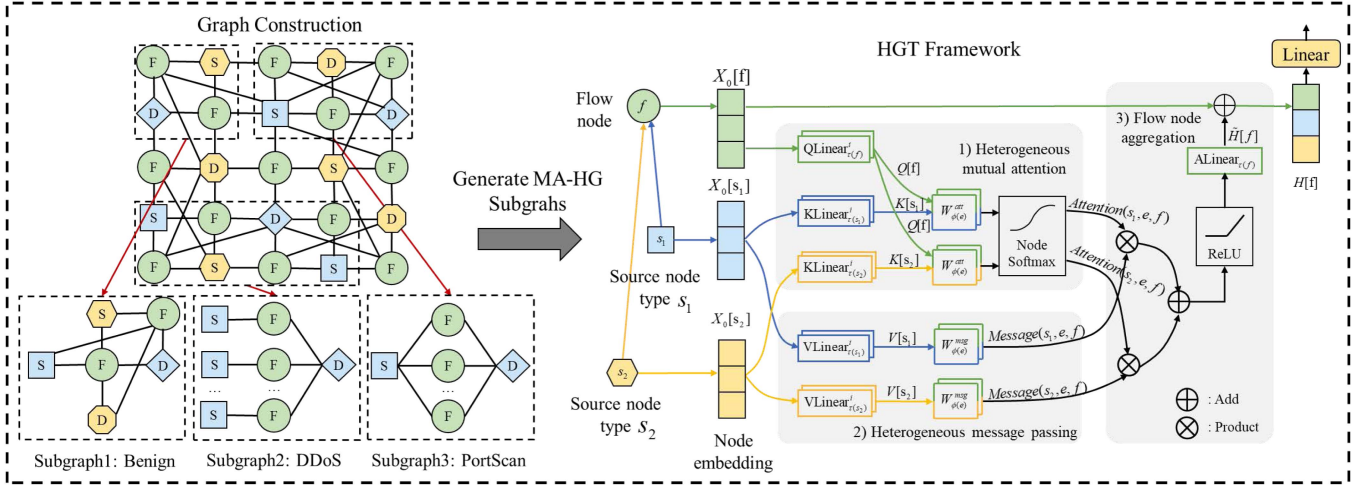


Fig. 4. Framework of MA-HGT.

flow data and encompasses all its statistical features. Thus, to acquire the network topological features of each flow data while preserving its statistical features, we can employ message passing and aggregation among *Flow* nodes to obtain the graph embedding for each *Flow* node. Lastly, by classifying all *Flow* nodes, we can determine the category of each corresponding network flow data.

C. Heterogeneous Graph Embedding

To mine implicit meta relationships between flows and their attributes in the graph construction, we use HGT [17] to get graph embeddings for all nodes, as shown in Fig. 4. The input for HGT is the heterogeneous graph G_I or G_E . For the sake of simplicity in mathematical notation, we will use the symbol X_0 to represent either of these graphs. Moreover, the outputs of HGT are the graph embeddings representing flows and their attributes, indicated as H .

Definition 3. Meta relation: We define $\langle \tau(s), \phi(e), \tau(t) \rangle$ as the meta relation for an edge $e = (s, t)$ linked from source node s to target node t .

There exists a one-to-one link between the *Flow* node in the constructed graph structure and the network flow statistics. Consequently, one can regard the graph embedding of the *Flow* node as the graph embedding of the network flow data. As a result, the *Flow* node is chosen as the target node, denoted as f , to reduce the computational complexity.

1) *Heterogeneous mutual attention:* Initially, we use three linear projection functions to transform node embeddings into the i th vector. In detail, the function $QLinear_{\tau(f)}^i$ is used to map the *Flow* node f into the i th query vector $Q^i(f)$, which has a dimension of $\mathbb{R}^d \rightarrow \mathbb{R}^{\frac{d}{h}}$. Here, d represents the dimension of $X_0[f]$, while $\frac{d}{h}$ denotes the vector dimension per head. Similarly, the functions $KLinear_{\tau(s)}^i$ and $VLinear_{\tau(s)}^i$ are also used to map the source node s to the i th key vector $K^i(s)$ and the i th value vector $V^i(s)$. It is important to highlight that these functions are indexed based on the node's type [37]. This implies that each kind of node has a distinct linear projection that aims to capture

the distribution disparities effectively.

$$Q^i(f) = QLinear_{\tau(f)}^i(X_0[f]) \quad (1)$$

$$K^i(s) = KLinear_{\tau(s)}^i(X_0[s]) \quad (2)$$

$$V^i(s) = VLinear_{\tau(s)}^i(X_0[s]) \quad (3)$$

Subsequently, it is necessary to compute the attention heads for a pair of nodes. In the process of computing the similarity between the Query vector $Q^i(f)$ and Key vector $K^i(s)$, a unique matrix $W_{\phi(e)}^{ATT} \in \mathbb{R}^{\frac{d}{h} \times \frac{d}{h}}$ is utilized for each specific edge type $\phi(e)$. By doing so, the model has the ability to capture various semantic relations, even when considering node-type pairs that are identical. Additionally, it is important to note that not all relationships have an equal impact on the target nodes. To account for this, we introduce a prior tensor $\mu \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{R}| \times |\mathcal{A}|}$ which represents the overall significance of each meta relation triplet. This tensor serves as an adaptive scaling factor for the attention mechanism. Hence, the calculation of $head_{att}^i(s, e, f)$ for edge $e = (s, f)$ is performed as follows:

$$head_{att}^i(s, e, f) = \left(K^i(s) W_{\phi(e)}^{att} Q^i(f)^T \right) \cdot \frac{\mu(\tau(s), \phi(e), \tau(f))}{\sqrt{d}} \quad (4)$$

Ultimately, the attention heads are concatenated, resulting in the generation of an attention vector for each pair of nodes. Next, we proceed to collect attention vectors from the neighboring nodes $N(f)$ for each target node f and apply the softmax function. Thus, the computation of the h -head attention for edge $e = (s, f)$ is performed as follows:

$$Attention(s, e, f) = \text{Softmax}_{\forall s \in N(f)} \left(\parallel_{i \in [1, h]} head_{att}^i(s, e, f) \right) \quad (5)$$

2) *Heterogeneous message passing:* First, the message head $head_{msg}^i(s, e, f)$ for edge $e = (s, f)$ is calculated by:

$$head_{msg}^i(s, e, f) = V^i(s) W_{\phi(e)}^{msg} \quad (6)$$

where $\mathbf{W}_{\phi(e)}^{\text{msg}} \in \mathbb{R}^{\frac{d}{h} \times \frac{d}{h}}$ is utilized to incorporate the edge dependence, analogous to the utilization of $\mathbf{W}_{\phi(e)}^{\text{ATT}}$. Then, we concatenate all the message heads to get the $\text{Message}(\mathbf{s}, \mathbf{e}, \mathbf{f})$ for edge $e = (s, f)$:

$$\text{Message}(\mathbf{s}, \mathbf{e}, \mathbf{f}) = \parallel_{i \in [1, h]} \text{head}_{\text{msg}}^i(\mathbf{s}, \mathbf{e}, \mathbf{f}) \quad (7)$$

3) *Flow node aggregation*: The $\text{Attention}(\mathbf{s}, \mathbf{e}, \mathbf{f})$ is regarded as the weight to average the corresponding $\text{Message}(\mathbf{s}, \mathbf{e}, \mathbf{f})$. The aforementioned action aggregates information from various source nodes and transfers it to the destination node f . As a result of this process, the vector $\tilde{\mathbf{H}}[\mathbf{f}]$ is updated as:

$$\tilde{\mathbf{H}}[\mathbf{f}] = \bigoplus_{\forall s \in N(f)} (\text{Attention}(\mathbf{s}, \mathbf{e}, \mathbf{f}) \cdot \text{Message}(\mathbf{s}, \mathbf{e}, \mathbf{f})) \quad (8)$$

Then, the *ReLU* activation function and linear projection ALinear are applied to the $\tilde{\mathbf{H}}[\mathbf{f}]$, and this is followed by a residual connection as:

$$\mathbf{H}[\mathbf{f}] = \text{ALinear}_{\tau(f)} \left(\text{ReLU}(\tilde{\mathbf{H}}[\mathbf{f}]) \right) + \mathbf{X}_0[\mathbf{f}] \quad (9)$$

Employing this approach, we obtain the output of a highly contextualized representation for the *Flow* node f . Subsequently, this representation is utilized as input for a single-layer fully connected network to perform flow node classification.

D. Attack Classification

In this step, the MA-HGT feeds the graph embedding of flow nodes $\mathbf{H}[\mathbf{f}]$ into a single-layer fully connected network to generate the system's output. Specifically, the quantity of neurons within the fully connected network is contingent upon the number of classification categories. We use the softmax activation function, calculated as follows:

$$\mathbf{Y}_f = \text{Softmax}(\mathbf{W} * \mathbf{H}[\mathbf{f}] + \mathbf{b}) \quad (10)$$

where $\mathbf{Y}_f$ is the predicted label of network flow data, i.e., the result of the MA-HGT, $\mathbf{W}$ is a weighted matrix and $\mathbf{b}$ is bias vector. We chose a single-layer fully connected network because it primarily serves as a dimensional transformation component in our framework, converting high-dimensional feature representations from the HGT into a format suitable for final classification tasks (binary or multi-class). The proposed MA-HGT's representational power allows this simple structure to effectively map graph features to classification space while minimizing overfitting risk from excessive parameters. For the output layer, we adopted the standard softmax activation function as it provides sufficient probabilistic differentiation to distinguish between benign traffic and various attack types.

E. Training on Subgraphs

To facilitate the training on giant heterogeneous graphs composed of network flows with hundreds of thousands of nodes and edges, we employ a method called MA-HGSampling. This method allows for the generation of subgraphs from the original gigantic graph. To begin, we include all direct neighbors $s \in N(f)$ of the *Flow* nodes in the budget $B[f]$, resulting in

TABLE III
CLASS LABEL, AND SIZE OF THE CAR-HACKING DATASET

Class Label	Number of Samples	Validation Instance	Test Instance	Attack Type
BENIGN	12243651	2448730	2448730	-
Dos attack	587521	117504	117504	Dos
Fuzzy attack	156085	31217	31217	Fuzzy
Spoofing the drive gear	523629	104726	104726	Spoofing
Spoofing the RPM gauge	654897	130979	130979	

$B[f][s]$. Subsequently, we calculate the sampling probabilities for these nodes:

$$P[f][s] = \frac{B[f][s]^2}{\|B[f]\|_2^2}, \quad (11)$$

where $B[\tau][s]$ is updated by the normalization degree of the candidate node concerning the *Flow* node:

$$B[f][s] = B[f][s] + \frac{1}{\text{len}(\text{Adj}[f])}, \quad (12)$$

where $\text{len}(\text{Adj}[f])$ is the length of adjacency matrix for each $\langle \tau(s), \phi(e), \tau(f) \rangle$ meta relation.

Subsequently, a specific number of nodes were sampled based on the probability P among all *Flow* nodes and the budget was allocated for each category of node. This iterative procedure is executed four times to generate a graph sample with a depth of four. Finally, the adjacency matrix between the sampled nodes is reconstructed. The MA-HGSampling technique ensures that the subgraph generated has a comparable number of nodes for each type as the original graph. Additionally, the subgraph is sufficiently dense to facilitate MA-HGT training on large-scale heterogeneous networks representing network traffic.

V. PERFORMANCE EVALUATION

A. Experimental Setup

The experiments utilize the PyTorch 2.0 library [38] and the PyTorch Geometric library [39] for model development. Furthermore, the trials were conducted on a workstation equipped with Ubuntu 18.04, an Intel Xeon Gold 6230R CPU, and an NVIDIA GeForce RTX 3090 GPU.

B. Data Description

This paper utilizes the following three IVN datasets to evaluate the proposed IDS:

- **CAR-HACKING**: The dataset was generated by capturing CAN traffic through the OBD-II port of an actual car during the execution of CAN attacks [33], [40]. The various attack types present in the CAR-HACKING dataset are displayed in Table III. The CAR-HACKING dataset encompasses several key attributes, namely the timestamp, CAN ID, data length code (DLC), and the DATA[0]-[7]. These attributes are derived from the arbitration field, control field, and data field of the CAN packets which are illustrates in Fig. 1 and Table I. Additionally, Table III displays the label

distributions of the original dataset, the training dataset, and the validation dataset.

- **CAN-INTRUSION:** This dataset mainly consists of impersonation attacks and injection attacks against the CAN bus [41]. In particular, impersonation attacks assume that the attacker is able to make physical modifications and can control the target node, stop the message transmission, and have the containing node send a message to replace the terminating node. On the other hand, there are two main different types of injection attacks in datasets: DOS and fuzzing attacks.
- **CICIoV2024:** This dataset was used by researchers to conduct five attacks on the complete interior of a 2019 Ford vehicle, including all ECU, over the CAN bus [42]. This dataset focuses on two types of threats against the CAN bus: DOS attacks and spoofing attacks. For DOS attacks, a malicious CAN user can change the arbitration value to initiate an attack. On the other hand, spoofing attacks include steering wheel spoofing, RPM spoofing, speed spoofing, and gas control spoofing.

In the process of developing the IDS for the IVN, we have chosen to incorporate all of the aforementioned properties as input features for the model. In order to address the memory constraints of the computer utilized in the experiments and to assess the efficacy of the model under label-constrained circumstances, we employed random down-sampling techniques to preprocess the data. This enabled the successful implementation of the proposed model for both training and testing purposes.

The scarcity of publicly available EVN benchmark datasets can be attributed to concerns about privacy and commercialisation [14], [43]. In contrast, Wireless Local Area Networks (WLANs) and cellular networks are frequently employed communication methods for IoV communication. Consequently, the attack on the conventional computer network can be regarded as analogous to the intrusion on the EVN [13], [28], [44]. Consequently, numerous research endeavors [13], [45], [46], [47], [48], [49], [50] have been undertaken to construct external IDS for vehicle networks. These projects utilize established intrusion detection datasets, such as ISCXID2012 [51] and UNSW-NB15 [52], to facilitate their development. Therefore, this paper utilizes the following three similar datasets to evaluate the proposed IDS for the performance on the EVN:

- **CIC-IDS2017:** This dataset is considered to be highly advanced due to its comprehensive range of features and inclusion of the up-to-date sorts of attacks [44], [53]. The datasets have a strong resemblance to authentic real-world data. The complete set of network traffic analysis results from the dataset was utilised by our IDS, encompassing over 80 network traffic attributes such as source and destination IPs, source and destination ports, and protocols. Table II presents several significant attributes of the dataset. Furthermore, this study establishes a correlation between each attack included in the CIC-IDS2017 dataset and the potential attacks discussed in Section III. The details pertaining to the CIC-IDS2017 dataset and the results of the correlation analysis are presented in Table IV.

TABLE IV
CLASS LABEL, AND SIZE OF THE CIC-IDS2017 DATASET

Class Label	Number of Samples	Validation Instance	Test Instance	Attack Type
BENIGN	2271320	454264	454264	-
Bot	1956	391	391	Botnets
DDoS	128025	25605	25605	DoS
DoS Hulk	230124	46025	46025	
DoS GoldenEye	10293	2059	2059	
DoS slowloris	5796	1159	1159	
DoS Slowhttptest	5499	1100	1100	
Heartbleed	11	2	2	Sniffing
PortScan	158804	31761	31761	
SSH-Patator	5897	1179	1179	
FTP-Patator	7935	1587	1587	Brute-Force
Infiltration	36	7	7	Infiltration
Web Attack-Brute Force	1507	301	301	Web-attack
Web Attack-XSS	652	130	130	
Web Attack-Sql Injection	21	4	4	

- **TON-IoT:** The TON-IoT dataset consists of the original dataset, the processed dataset, and the training test dataset [54]. In this paper, we select the training test dataset from the TON-IoT dataset. This dataset contains a total of nine attacks, namely Backdoor Attack, DoS Attack, DDoS Attack, Injection Attack, Password Attack, Ransomware, Scanning, xss Attack and Man-in-the-Middle Attack. The training test dataset contains 42 network traffic attributes such as source and destination IPs, source and destination ports, and protocols.
- **CIC-UNSW-NB15:** The CIC-UNSW-NB15 dataset uses the CICFlowMeter network traffic analysis tool to extract a new set of features from the captured network traffic data provided by UNSW-NB15, including a total of 83 features such as source and destination IPs, source and destination ports, and protocols [55]. This dataset contains a total of nine types of attacks, Fuzzer, Analytics, Backdoor, DoS, Vulnerability, Generic, Reconnaissance, Shellcode, and Worms.

The aforementioned attributes are utilised as input features in the proposed model, allowing it to optimise the learning of characteristics that enhance the performance of the proposed IDS. Furthermore, the analysis in Tables III and IV demonstrates that the CAR-HACKING and CIC-IDS2017 datasets comprehensively encompass the primary threat categories encountered in IoV environments. These datasets are widely adopted within the field of connected vehicle security research and possess high representativeness [13], [14], [28], [35], [45]. Therefore, a two-tiered experimental approach was adopted: (1) Comprehensive evaluation employs all six datasets for foundational experiments to demonstrate the proposed model's generalization capability across diverse network environments and attack scenarios. (2) Representative evaluation uses CAR-HACKING and CIC-IDS2017 as representative IVN and EVN datasets for specialized analyses of specific scenarios or dimensions.

C. Comparative Methods

In this study, we conducted a comparative analysis of the proposed IDS against eight approaches. The selection of machine learning or deep learning-based IDS baselines follows the following criteria: (1) Methods employing classical machine learning or deep learning algorithms that are widely cited in academia or serve as comparative benchmarks. (2) Whether the method accounts for the characteristics of IVN or EVN, or is specifically designed for these environments. (3) Whether the ACC and FAR meet the fundamental requirements for IoV security applications.

- **Rule-based IDS:** This rule-based IDS integrates multiple rule types, including statistical threshold rules [56], attack pattern rules [57], domain expert rules [32] and state-related rules [58]. We integrated multiple rules to balance method generality with detection capability. This design enables the IDS to adapt to diverse environments such as IVN and EVN, while delivering robust detection performance without requiring machine learning models.
- **AdaBoost-based IDS [59]:** AdaBoost is an iterative ensemble learning technique that constructs a robust classifier by aggregating multiple weak classifiers to enhance the accuracy of network traffic classification.
- **Gradient Boosting Decision Trees (GBDT)-based IDS [60]:** GBDT involves each iteration's iterative training of decision trees to rectify the faults made in the preceding iteration to classify network traffic effectively.
- **Random Forest(RF)-based IDS [35]:** RF enhances the performance and stability of intrusion detection models by aggregating the predicted results through voting or averaging.
- **Deep Neural Networks(DNN)-based IDS [44]:** DNN is a common variant of feed-forward neural networks with six hidden layers.
- **1D Convolutional Neural Network(1DCNN)-based IDS [61].** The 1DCNN extracts network traffic features through four 1D convolutional layers and 1D max pooling and classifies them through a single-layer neural network.
- **Long Short-Term Memory(LSTM)-based IDS [62].** LSTM can efficiently capture long-term relationships in network stream data, making it well-suited for intrusion detection tasks. LSTM contains two LSTM layers, one 1D adaptive average pooling layer, one dropout layer and two single-layer neural networks.
- **Gated Recurrent Unit(GRU)-based IDS [63].** GRU incorporates a gating mechanism that allows the GRU to efficiently capture long-term dependencies in streaming data and mitigate the issue of gradient vanishing. The architecture comprises two GRU layers followed by a simple one-layer neural network.

For all of the comparison methods used in the experiments, we referenced the parameter settings recommended in the original literature where these methods were proposed and optimally tuned these parameters to the characteristics of the datasets we used. Specifically, we tuned the grid search parameters for each algorithm to ensure that they would perform optimally on the datasets.

D. Validation Metrics

To assess the effectiveness of the proposed model, we employ several assessment measures to gauge its classification performance. These metrics include Accuracy (ACC), DR, FAR, and F1-Score. The ACC metric is used to evaluate the overall performance of the proposed model. The DR measures the ability of the given model to correctly identify instances belonging to a specific category. The F1-Score is a metric that combines the precision rate and recall rate into a single value using the harmonic mean. The metrics are defined in the following manner:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

$$DR = \frac{TP}{TP + FN}$$

$$FAR = \frac{FP}{TN + FP}$$

$$F1 - Score = \frac{2 \times TP}{2 \times TP + FP + FN}$$

where the variables TP, TN, FP, and FN represent the quantities of true positive, true negative, false positive, and false negative, respectively.

E. Performance Results

1) *Performance of binary classification:* To assess the effectiveness of the proposed IDS for intrusion detection, the MA-HGT algorithm and eight state-of-the-art baseline approaches are developed. These methods are trained and tested on three IVN and EVN distinct data sets, which accurately represent internal and external vehicle network traffic data, respectively. All comparative approaches have demonstrated favourable performance for intrusion detection in IVN or EVN. The results of the experiment for all methods are shown in Fig. 5. For accuracy metric, MA-HGT achieves 99.99% , 99.95% and 99.97% on the CAR-HACKING, CAN-INTRUSION and CICIoV2024, respectively. The accuracy of MA-HGT on the IVN datasets is at least 3.73% , 15.55% , 3.45% higher than the other eight methods. On the other hand, MA-HGT achieves 99.98% , 99.95% and 99.99% accuracy on the CIC-IDS2017, TON-IoT and CIC-UNSW-NB15, respectively. The accuracy of MA-HGT on the EVN datasets is at least 3.51% , 13.96% , 2.67% higher than the other eight methods. Meanwhile, as depicted in Fig. 5, the result demonstrates that MA-HGT attains F1-Scores of 99.99% , 99.97% and 99.70% on the CAR-HACKING, CAN-INTRUSION and CICIoV2024, respectively. The F1-Score of MA-HGT on the IVN datasets is at least 13.29% , 9.02% , 2.42% higher than those achieved by the other eight methods. Additionally, MA-HGT achieves F1-Scores of 99.88% , 99.97% and 99.99% F1-Score on the CIC-IDS2017, TON-IoT and CIC-UNSW-NB15, respectively. The MA-HGT approach demonstrates a minimum increase in F1-Score of 9.00% , 3.71% and 4.40% compared to the other eight methods, as observed in the EVN datasets. Besides, upon observing the additional two indices shown in Tables V and VI, it becomes

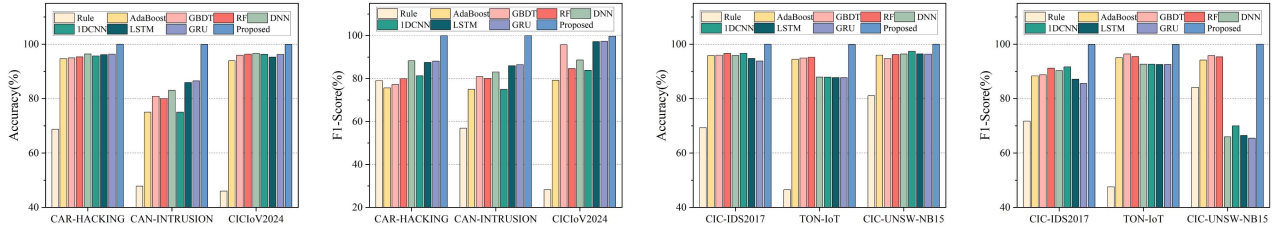


Fig. 5. Binary classification performance of all methods on IVN and EVN datasets.

TABLE V
DETECTION RATE AND FALSE ALARM RATE OF ALL METHODS ON THREE IVN DATASETS

Method	CAR-HACKING		CAN-INTRUSION		CICIoV2024	
	DR(%)	FAR(%)	DR(%)	FAR(%)	DR(%)	FAR(%)
Rule	73.41	49.88	45.86	46.40	81.00	59.28
AdaBoost	61.33	0.13	98.00	90.43	89.04	5.40
GBDT	63.12	0.04	97.53	75.08	91.77	5.00
RF	68.31	0.43	97.43	78.26	76.77	4.70
DNN	99.94	4.16	95.05	58.94	96.99	4.86
1DCNN	69.05	0.14	97.99	90.00	73.17	3.20
LSTM	99.68	4.42	97.19	53.86	96.99	2.88
GRU	99.94	4.24	97.40	52.19	96.99	2.82
Proposed	99.99	0.02	99.96	0.11	99.99	0.08

TABLE VI
DETECTION RATE AND FALSE ALARM RATE OF ALL METHODS ON THREE EVN DATASETS

Method	CIC-IDS2017		TON-IoT		CIC-UNSW-NB15	
	DR(%)	FAR(%)	DR(%)	FAR(%)	DR(%)	FAR(%)
Rule	77.95	39.43	31.74	6.08	95.23	34.88
AdaBoost	80.85	2.55	97.24	26.12	95.49	3.62
GBDT	80.57	2.25	97.78	8.05	97.91	2.66
RF	89.54	1.68	98.21	21.42	97.96	3.79
DNN	97.90	4.65	97.77	50.34	97.48	5.66
1DCNN	94.87	2.99	97.78	50.58	74.20	4.98
LSTM	90.88	4.33	97.80	51.17	97.87	4.61
GRU	93.06	6.00	97.77	51.13	97.20	4.70
Proposed	99.99	0.03	99.96	0.11	99.99	0.24

evident that MA-HGT exhibits a noticeable enhancement in detection performance compared to the baseline technique. This demonstrates that the proposed IDS effectively acquires a unified representation encompassing topological and statistical characteristics of network flow data. Furthermore, the MA-HGT model performs better than deep learning techniques such as 1DCNN and LSTM across all evaluation criteria, particularly when evaluated on the EVN datasets. One possible explanation for this phenomenon is the high feature dimensionality of the EVN datasets; for example, the CIC-IDS2017 dataset contains more than 80 feature dimensions. Meanwhile, the absence of a multi-attention mechanism in these approaches hinders their ability to effectively concentrate on important features, leading to suboptimal detection results.

Additionally, the Receiver Operating Characteristic (ROC) and Area Under the Curve (AUC) findings for all approaches on CAR-HACKING and CIC-IDS2017 datasets are presented in Fig. 6. It is evident that the AUC scores obtained by the MA-HGT approach surpass those of the other methods, exceeding 0.9999 on both datasets. Furthermore, Fig. 7 presents the Precision-Recall (PR) curves and Average Precision (AP) scores

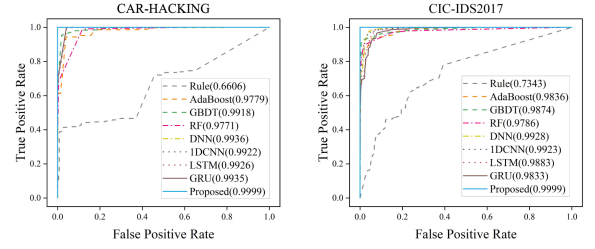


Fig. 6. ROC curves of all methods on two dataset.

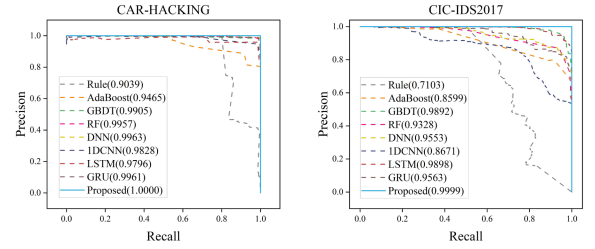


Fig. 7. PR curves of all methods on two dataset.

for all methods across the CAR-HACKING and CIC-IDS2017 datasets. The MA-HGT method achieved superior AP scores compared to all baseline approaches, exceeding 0.9999 on both datasets. The findings of these experiments suggest that the proposed MA-HGT has superior classification performance for IVN and EVN flow data compared to alternative approaches. The MA-HGT framework can transform traffic flows into a topological structure of heterogeneous graph data, which can then be effectively utilised for intrusion detection through graph embedding technology.

2) *Performance of multi-classification*: To evaluate the multi-classification performance of MA-HGT, we generated confusion matrices for the test sets of the IVN and EVN datasets, as depicted in Fig. 8, respectively.

For the CAN-HACKING dataset, MA-HGT can accurately detect three types of attacks: DoS, RPM spoofing and gear spoofing. The accuracy rate of MA-HGT was found to be exceptionally high, reaching 99.98%. This high accuracy can be attributed to the algorithm's ability to effectively identify and classify instances of DoS, RPM spoofing, and gear spoofing attacks. Moreover, in the case of Fuzzy attacks, which exhibit attack patterns that closely resemble those of other attacks, MA-HGT demonstrated minimal misclassifications. Furthermore, MA-HGT achieved high accuracy rates of 98.45% and 99.99% on the CAN-intrusion and CICIoV2024 datasets, respectively.

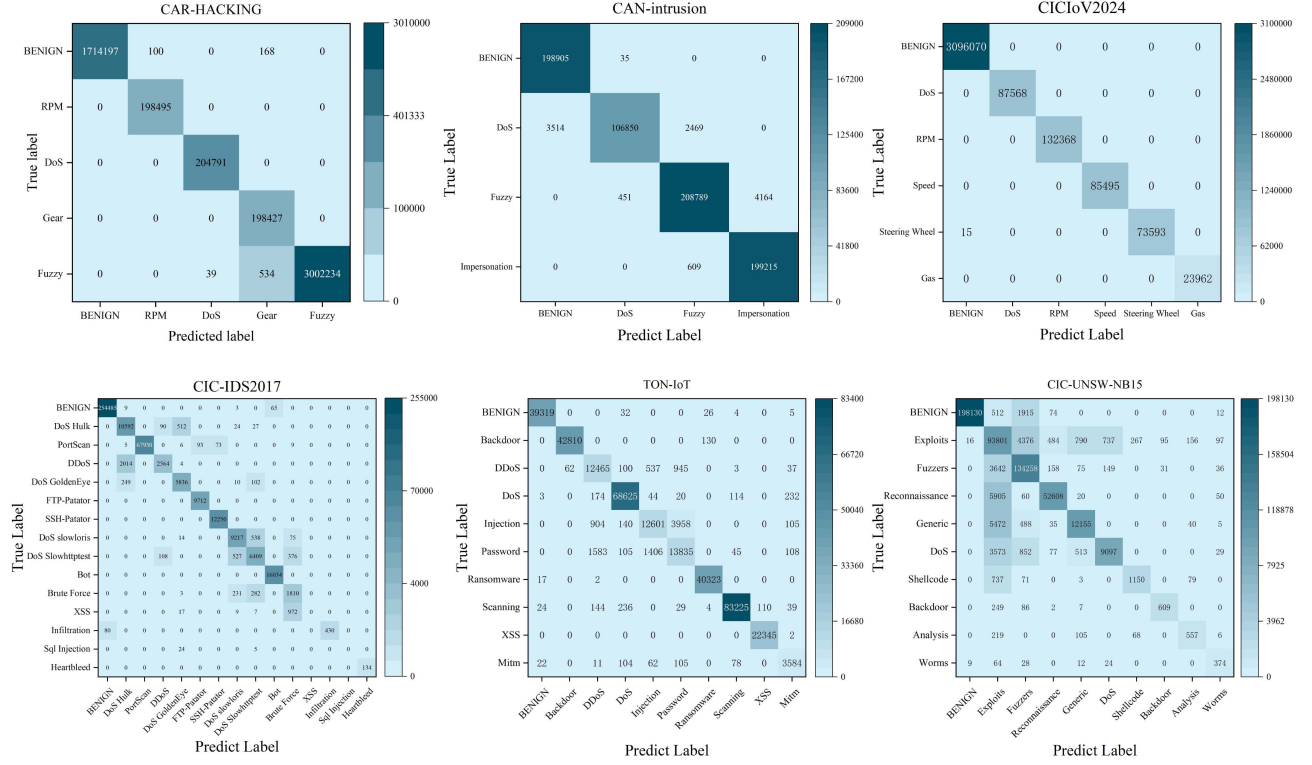


Fig. 8. Confusion matrix of proposed methods for the test set on IVN and EVN datasets.

Despite the presence of different attack types and complicated assault patterns within the CIC-IDS2017 dataset, the MA-HGT model demonstrates a remarkable accuracy rate of 98.38%. The MA-HGT model achieves an accuracy of 1 in identifying and classifying four common attack patterns: Bot, Heartbleed, FTP-Patator, and SSH-Patator attacks. However, the accuracy for various DoS attacks is slightly lower. This might be attributed to DoS attack patterns exhibiting a high degree of similarity. Moreover, the performance of MA-HGT in the context of Infiltration and XSS attacks is suboptimal. One potential explanation for this outcome is the limited number of samples available for the two attacks, impeding correct attack identification. Additionally, MA-HGT achieved high accuracy rates of 96.62% on the TON-IoT dataset and 93.92% on the CIC-UNSW-NB15 dataset. The relatively lower accuracy on the CIC-UNSW-NB15 dataset is attributed to the severe class imbalance in the data.

The factor contributing to the high-performance multi-classification of MA-HGT is the ability of each head within the HGT to acquire distinct representations concurrently. This enables the model to selectively attend to various locations or features concurrently, thereby enhancing the model's classification performance on intricate data.

3) *Ablation study:* To comprehensively evaluate the role of each MA-HGT module in intrusion detection, we conducted ablation experiments on the proposed system using both IVN and EVN datasets, with results presented in Fig. 9. The ablation study focuses on three key modules: the MA-HG construction method, the HGT module, and the MA-HGSampling method. We designed the following ablation experiments to systematically evaluate each module's contribution: (1) w/o HL: excludes

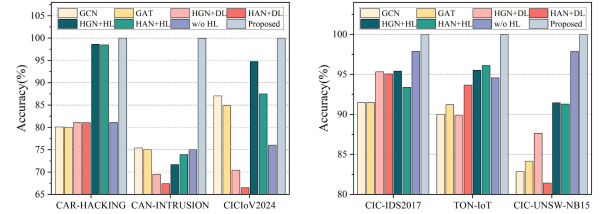


Fig. 9. Ablation study of the proposed IDS on IVN and EVN datasets.

the MA-HG Sampling method, utilizing only the ordinary Dataloader; (2) HAN+HL: replaces the HGT module with the Heterogeneous Graph Attention Network (HAN) [64]; (3) HGN+HL: substitutes the HGT module with a Heterogeneous Graph Convolutional Network (HGCN) [39]; (4) HAN+DL: combines the HAN with the ordinary Dataloader, omitting the multi-attribute heterogeneous graph sampling method; (5) HGN+DL: employs the HGCN with the ordinary Dataloader, excluding the MA-HG Sampling method; (6) GAT: utilizes homographs instead of heterogeneous graphs, implementing the Graph Attention Network (GAT) [65]; (7) GCN: employs homographs rather than heterogeneous graphs, applying the Graph Convolutional Network (GCN) [66].

The results of the ablation experiments revealed performance variations across different datasets. Overall, MA-HGT achieved the highest accuracy on each dataset compared to all ablation methods. These results demonstrate that MA-HGT provides efficient detection Performance, confirming the rational design of MA-HGT and the indispensable and significant contribution of

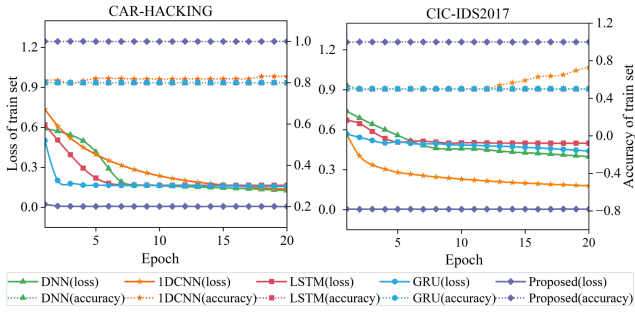


Fig. 10. Loss and accuracy of five methods on two datasets during training.

each module to its detection capability. Furthermore, the relative importance of each module varies across different datasets. For example, the MA-HGSampling method plays a more crucial role on CAR-HACKING, CICIoV2024, and TON-IoT datasets, whereas the HGT module contributes more significantly to performance on CIC-IDS2017 and CIC-UNSW-NB15. On the CAR-INTRUSION dataset, the ablation results indicate that each module contributes equally to the overall performance. Notably, the model's accuracy decreases significantly when the common homography construction method is used instead of our proposed heterogeneous graph construction approach.

4) *Training efficiency of MA-HGT*: In this experiment, we will validate the training efficiency of MA-HGT at a limited labeling rate. We conducted training on MA-HGT and four different deep-learning methods for a total of 20 epochs. The training data utilised had a labelling rate of 80%. Based on the information shown in Fig. 10, it is evident that the loss and accuracy of MA-HGT exhibit stabilisation after a single epoch across both datasets. This suggests that MA-HGT achieves convergence and optimal performance after a single training epoch. Furthermore, the stability of loss and accuracy from epochs 2-20 further support this conclusion. Regarding the remaining four deep learning methods, it can be observed that even after 20 epochs of training, the loss and accuracy exhibit a fluctuating pattern, indicating a lack of convergence. This observation suggests that the training efficiency of the proposed MA-HGT model surpasses that of several existing deep learning methods. One potential rationale is that the heterogeneous graph representation in MA-HGT enables the effective depiction of both the topological relationships and statistical characteristics within the network flow data. Simultaneously, incorporating a multi-head attention mechanism facilitates the rapid identification of crucial features pertaining to normal and abnormal network activities within the heterogeneous graph constructed through this representation. Consequently, MA-HGT exhibits swift convergence and efficient detection capabilities, requiring only a minimal number of epochs.

In order to provide additional evidence of the training efficacy of MA-HGT, the alterations in detection rate and F1 score over the course of 20 training epochs for both MA-HGT and the four deep learning techniques on the CAR-HACKING and CIC-IDS2017 datasets are depicted in Figs. 11 and 12. These figures illustrate the changes in detection rate, false alarm rate, and F1-Score throughout the training process of the two datasets.

During the training process using the aforementioned datasets, it was observed that the curves representing the three metrics of MA-HGT reached their highest values at the completion of one epoch. In contrast, compared to MA-HGT, the convergence speeds of the three metric curves in the remaining four deep learning methods are comparatively slower, and their convergence effects are less pronounced. It is worth mentioning that the three metric curves associated with the IDCNN approach exhibit persistent oscillations in the 20 epochs. The observed phenomenon can be attributed to the high sensitivity of IDCNN to unbalanced network data, which makes it unable to solve the classification problem with unbalanced sample categories effectively. Furthermore, experimental results show that MA-HGT has high performance in classification on both IVN and EVN flow data. The MA-HGT converges quickly and can effectively handle datasets with label-limited datasets.

5) *Effect of labeling rate*: To assess the performance of MA-HGT under varying labelling rates of the dataset, the accuracy of MA-HGT and seven learning-based methods was assessed at various labelling rates spanning from 1% to 100% on the CAR-HACKING and CIC-IDS2017 datasets. Based on the analysis of Fig. 13, it is evident that most approaches exhibit a declining pattern in accuracy as the labelling rate declines. This observation suggests that the proportion of labelled data in the training set plays a significant role in influencing the accuracy of IDS. As depicted in Fig. 13, it is evident that the accuracy rate of MA-HGT consistently surpasses that of the other baseline methods. Furthermore, there is minimal variation in the accuracy rate of MA-HGT as the labelling rate declines from 100% to 1%. The heterogeneous graph construction method's efficacy and multi-head attention mechanisms in MA-HGT have been demonstrated to enhance the model's ability to concentrate on critical features, resulting in improved classification performance. Furthermore, the experimental findings also demonstrate that integrating topological and statistical traffic flow characteristics in MA-HGT enhances its ability to capture the underlying representations of network traffic flow relationships. As a result, it achieves outstanding classification performance across various label rates. In addition, the commendable performance exhibited by MA-HGT serves as evidence that the issue of inadequate learnable information inside the model may be effectively addressed through the construction of a heterogeneous graph topology for network flow data, coupled with the utilisation of HGT for graph embedding, even when faced with limitations in labelling rate.

6) *Real-time performance, model complexity, and computational efficiency*: To assess the feasibility of deploying the proposed IDS in real-world vehicular environments, we conducted five independent experiments to evaluate different aspects of real-time performance, model complexity, and computational efficiency. Each experiment was designed to measure a specific performance metric: (1) average test time, (2) model size, (3) total number of parameters, (4) Floating Point Operations Per Second (FLOPs) and (5) average power consumption. All experiments were conducted using the hardware environment described in Section V-A and the results are presented in Fig. 14(a). The experimental results demonstrate that the

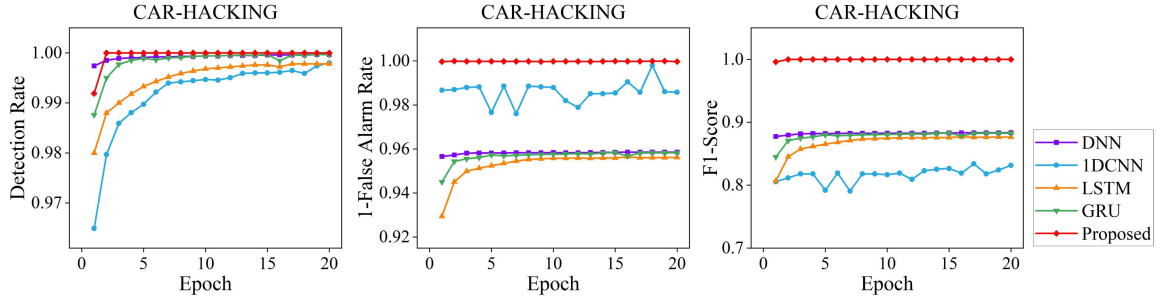


Fig. 11. Detection rate, false alarm rate, F1-Score of five methods on CAR-HACKING dataset during training.

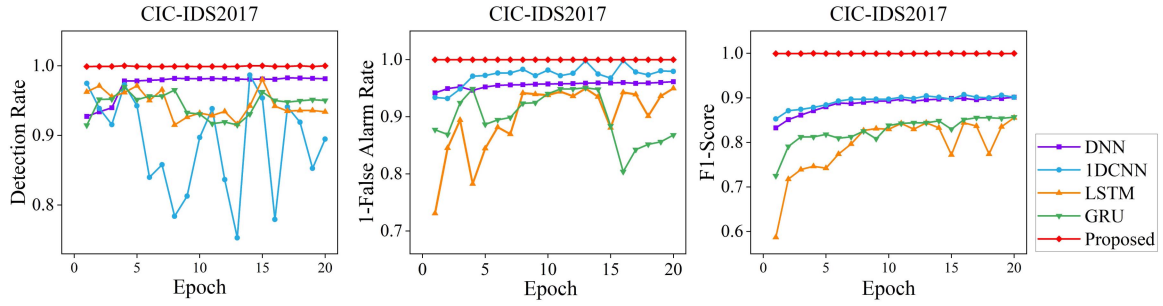


Fig. 12. Detection rate, false alarm rate, F1-Score of five methods on CIC-IDS2017 dataset during training.

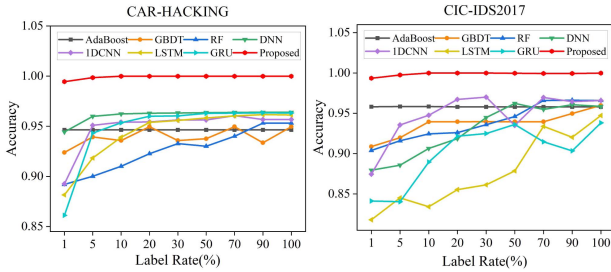


Fig. 13. Accuracy of five methods on two datasets with different labeling rates.

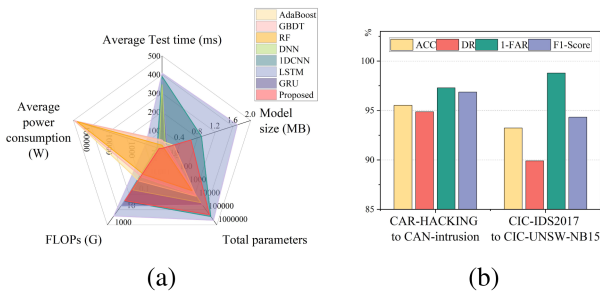


Fig. 14. (a) Comparison of real-time performance, model complexity, and computational efficiency and (b) Cross-dataset intrusion detection experiment results.

proposed method achieves the shortest average test time, with its inference speed outperforming all baseline methods across all six datasets. Additionally, the model size, total number of parameters, FLOPs and average power consumption of the proposed IDS remain at relatively low to moderate levels compared

to the baseline methods. To further demonstrate the feasibility of deploying One² in automotive applications, we conducted an analysis mapping experimental results from workstation to real-vehicle environments. According to [67], the latency for safety services in vehicular environments should be below 100ms. Concurrently, detection latency should be minimized to prevent interference with normal vehicle operation [68]. The proposed model achieved a GPU test time of 5.13ms. However, CPU inference times in practical tests were approximately 2.5x to 14.9x times longer than GPU times [69]. We employed a conservative 45x for estimation. Furthermore, the introduction of INT8 quantization [70] and model pruning techniques [71] can respectively enhance inference speed by 3.8x to 4.7x and 6.67x (with only a 1% impact on model accuracy). Using a conservative estimate of 9.11ms for CPU inference time, this falls below the latency requirement for vehicular network safety services. Furthermore, our model remains sufficiently compact at 0.65MB, which is considerably less than the memory capacity of automotive computing systems, which typically possess over 1GB of RAM [14]. Therefore, experimental results demonstrate that the proposed method exhibits superior feasibility for deployment in real vehicular systems.

7) *Cross-domain detection*: To demonstrate the robustness of the proposed method against unseen attack scenarios, we conducted cross-dataset intrusion detection experiments. We performed two sets of cross-dataset experiments: training on CAR-HACKING and testing on CAN-intrusion datasets, and training on CIC-IDS2017 and testing on CIC-UNSW-NB15 datasets. These dataset pairs were selected because they share similar network traffic characteristics, making them appropriate for cross-dataset validation. Fig. 14(b) demonstrates that the

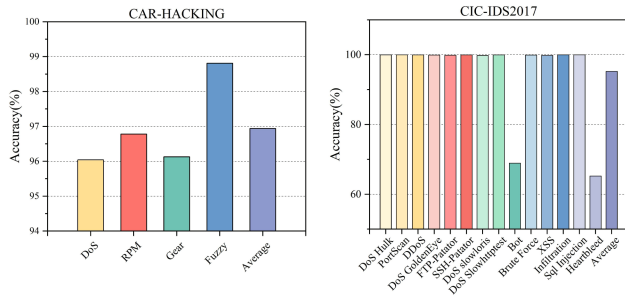


Fig. 15. Zero-day attack detection test results of proposed methods on two datasets.

proposed method achieved consistently high performance across all four evaluation metrics in both cross-dataset experiments. These results confirm that the proposed method demonstrates strong adaptability to unseen scenarios.

8) *Zero-day attack detection*: To investigate the detection performance of the proposed IDS against zero-day attacks, we conducted zero-day attack evaluation experiments using the CAR-HACKING and CIC-IDS2017 datasets [53]. In each experiment, we designated one attack type from the dataset as the zero-day attack, extracted all corresponding attack instances, and combined them with a proportionate amount of randomly sampled benign data to form the test set. The training set consisted of all remaining data from the original dataset. This approach ensured the complete absence of zero-day attack patterns in the training data. We conducted these experiments for each attack type present in both the CAR-HACKING and CIC-IDS2017 datasets. The results are presented in Fig. 15. The experimental results indicate that the proposed IDS achieves accuracy exceeding 90% for most zero-day attack types across both datasets, confirming its effectiveness in detecting zero-day attack patterns.

VI. CONCLUSION

In this paper, we propose One² IDS for label-limited IoV. Initially, a graph construction is developed to represent the topology of network traffic. Subsequently, the MA-HGT is employed to obtain graph embeddings. Finally, the node classification is performed on the graph for intrusion detection. The performance of the MA-HGT is evaluated using three IVN and EVN datasets, respectively. The experimental results demonstrate that MA-HGT exhibits superior classification performance in binary and multi-class classification. Moreover, MA-HGT demonstrates notable convergence speed. Furthermore, despite a limited labeling rate, the MA-HGT model achieves exceptional detection accuracy, surpassing other state-of-the-art methodologies. A critical direction for future work involves conducting comprehensive on-target evaluations on representative automotive hardware platforms (such as NVIDIA DRIVE, NXP S32) to validate real-time performance under authentic IVN and EVN loads. We will thoroughly analyze the complete pipeline timing, worst-case execution time bounds, batching/windowing effects on detection latency, and system coexistence with other safety-critical tasks following ASIL considerations—essential factors for successful integration into production vehicle environments.

REFERENCES

- [1] L. Yang, A. Moubayed, I. Hamieh, and A. Shami, "Tree-based intelligent intrusion detection system in Internet of Vehicles," in *Proc. 2019 IEEE Glob. Commun. Conf.*, 2019, pp. 1–6.
- [2] B. Ji et al., "Survey on the Internet of Vehicles: Network architectures and applications," *IEEE Commun. Standards Mag.*, vol. 4, no. 1, pp. 34–41, Mar. 2020.
- [3] C. Miller and C. Valasek, "Remote exploitation of an unaltered passenger vehicle," *Black Hat USA*, vol. 2015, no. S91, pp. 1–91, 2015.
- [4] S. Nie, L. Liu, Y. Du, and W. Zhang, "Over-the-air: How we remotely compromised the gateway, BCM, and autopilot ECUs of tesla cars," *Briefing, Black Hat USA*, vol. 91, pp. 1–19, 2018.
- [5] K. Kim, J. S. Kim, S. Jeong, J.-H. Park, and H. K. Kim, "Cybersecurity for autonomous vehicles: Review of attacks and defense," *Comput. Secur.*, vol. 103, 2021, Art. no. 102150.
- [6] B. Lampe and W. Meng, "A survey of deep learning-based intrusion detection in automotive applications," *Expert Syst. Appl.*, vol. 221, 2023, Art. no. 119771.
- [7] K. Ilgun, R. A. Kemmerer, and P. A. Porras, "State transition analysis: A rule-based intrusion detection approach," *IEEE Trans. Softw. Eng.*, vol. 21, no. 3, pp. 181–199, Mar. 1995.
- [8] V. Kumar, D. Sinha, A. K. Das, S. C. Pandey, and R. T. Goswami, "An integrated rule based intrusion detection system: Analysis on UNSW-NB15 data set and the real time online dataset," *Cluster Comput.*, vol. 23, pp. 1397–1418, 2020.
- [9] S. Al-Sarawi, M. Anbar, B. A. Alabsi, M. A. Aladaileh, and S. D. A. Rihan, "Passive rule-based approach to detect sinkhole attack in RPL-based Internet of Things networks," *IEEE Access*, vol. 11, pp. 94081–94093, 2023.
- [10] P. Mishra, V. Varadharajan, U. Tupakula, and E. S. Pilli, "A detailed investigation and analysis of using machine learning techniques for intrusion detection," *IEEE Commun. Surveys Tut.*, vol. 21, no. 1, pp. 686–728, First Quarter 2019.
- [11] S. Sharma and A. Kaul, "A survey on intrusion detection systems and honeypot based proactive security mechanisms in VANETs and VANET cloud," *Veh. Commun.*, vol. 12, pp. 138–164, 2018.
- [12] R. Bitton and A. Shabtai, "A machine learning-based intrusion detection system for securing remote desktop connections to electronic flight bag servers," *IEEE Trans. Dependable Secur. Comput.*, vol. 18, no. 3, pp. 1164–1181, May/Jun. 2021.
- [13] J. Ashraf, A. D. Bakhshi, N. Moustafa, H. Khurshid, A. Javed, and A. Beheshti, "Novel deep learning-enabled LSTM autoencoder architecture for discovering anomalous events from intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4507–4518, Jul. 2021.
- [14] L. Yang, A. Moubayed, and A. Shami, "MTH-IDS: A multitiered hybrid intrusion detection system for Internet of Vehicles," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 616–632, Jan. 2022.
- [15] X. Deng, J. Zhu, X. Pei, L. Zhang, Z. Ling, and K. Xue, "Flow topology-based graph convolutional network for intrusion detection in label-limited IoT networks," *IEEE Trans. Netw. Serv. Manag.*, vol. 20, no. 1, pp. 684–696, Mar. 2023.
- [16] N. Chaabouni, M. Mosbah, A. Zemmari, C. Sauvignac, and P. Faruki, "Network intrusion detection for IoT security based on learning techniques," *IEEE Commun. Surveys Tut.*, vol. 21, no. 3, pp. 2671–2701, Third Quarter 2019.
- [17] Z. Hu, Y. Dong, K. Wang, and Y. Sun, "Heterogeneous graph transformer," in *Proc. Web Conf.*, 2020, pp. 2704–2710.
- [18] H. Yu, H. Zhang, Z. Zhang, and S. Yang, "State estimation of lithium-ion batteries via physics-machine learning combined methods: A methodological review and future perspectives," *eTransportation*, vol. 24, 2025, Art. no. 100420.
- [19] A. Alshammari, M. A. Zohdy, D. Debnath, and G. Corser, "Classification approach for intrusion detection in vehicle systems," *Wireless Eng. Technol.*, vol. 9, no. 4, pp. 79–94, 2018.
- [20] A. Derhab, M. Belaoued, I. Mohiuddin, F. Kurniawan, and M. K. Khan, "Histogram-based intrusion detection and filtering framework for secure and safe in-vehicle networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 3, pp. 2366–2379, Mar. 2022.
- [21] L. Yang, A. Shami, G. Stevens, and S. De Rusett, "LCCDE: A decision-based ensemble framework for intrusion detection in the Internet of Vehicles," in *Proc. IEEE Glob. Commun. Conf.*, 2022, pp. 3545–3550.
- [22] H. Yu et al., "State of charge estimation method by using a simplified electrochemical model in deep learning framework for lithium-ion batteries," *Energy*, vol. 278, 2023, Art. no. 127846.

- [23] E. Seo, H. M. Song, and H. K. Kim, "GIDS: GAN based intrusion detection system for in-vehicle network," in *Proc. 16th Annu. Conf. Privacy Secur. Trust*, 2018, pp. 1–6.
- [24] T. P. Nguyen, H. Nam, and D. Kim, "Transformer-based attention network for in-vehicle intrusion detection," *IEEE Access*, vol. 11, pp. 55389–55403, 2023.
- [25] X. Han et al., "STIDM: A spatial and temporal aware intrusion detection model," in *Proc. IEEE Int. Conf. Trust Secur. Privacy Comput. Commun.*, 2020, pp. 370–377.
- [26] J. Zhao, X. Liu, Q. Yan, B. Li, M. Shao, and H. Peng, "Multi-attributed heterogeneous graph convolutional network for bot detection," *Inf. Sci.*, vol. 537, pp. 380–393, 2020.
- [27] Y. Cao, H. Jiang, Y. Deng, J. Wu, P. Zhou, and W. Luo, "Detecting and mitigating DDoS attacks in SDN using spatial-temporal graph convolutional network," *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 6, pp. 3855–3872, Nov./Dec. 2022.
- [28] L. Yang and A. Shami, "A transfer learning and optimized CNN based intrusion detection system for Internet of Vehicles," in *Proc. IEEE Int. Conf. Commun.*, 2022, pp. 2774–2779.
- [29] H. Liang et al., "Network and system level security in connected vehicle applications," in *Proc. IEEE/ACM Int. Conf. Comput.-Aided Des.*, 2018, pp. 1–7.
- [30] W. Wu et al., "A survey of intrusion detection for in-vehicle networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 3, pp. 919–933, Mar. 2020.
- [31] J. Liu, S. Zhang, W. Sun, and Y. Shi, "In-vehicle network attacks and countermeasures: Challenges and future directions," *IEEE Netw.*, vol. 31, no. 5, pp. 50–58, 2017.
- [32] S.-F. Lokman, A. T. Othman, and M.-H. Abu-Bakar, "Intrusion detection system for automotive controller area network (CAN) bus system: A review," *EURASIP J. Wireless Commun. Netw.*, vol. 2019, pp. 1–17, 2019.
- [33] E. Seo, H. M. Song, and H. K. Kim, "GIDS: GAN based intrusion detection system for in-vehicle network," in *Proc. 16th Annu. Conf. Privacy Secur. Trust*, 2018, pp. 1–6.
- [34] H. Jiang, Y. Ren, Y. Zhao, Z. Cui, and H. Yu, "Toward city-scale vehicular crowd sensing: A decentralized framework for online participant recruitment," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 17800–17813, Oct. 2025.
- [35] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. Int. Conf. Inf. Syst. Secur. Privacy*, 2018, pp. 108–116.
- [36] B. Groza and P.-S. Murvay, "Efficient intrusion detection with bloom filtering in controller area networks," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 4, pp. 1037–1051, Apr. 2019.
- [37] A. Ma et al., "Single-cell biological network inference using a heterogeneous graph transformer," *Nat. Commun.*, vol. 14, no. 1, 2023, Art. no. 964.
- [38] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 8026–8037.
- [39] M. Fey and J. E. Lenssen, "Fast graph representation learning with PyTorch geometric," in *Proc. ICLR Workshop Representation Learn. Graphs Manifolds*, 2019, pp. 1–9.
- [40] H. M. Song, J. Woo, and H. K. Kim, "In-vehicle network intrusion detection using deep convolutional neural network," *Veh. Commun.*, vol. 21, 2020, Art. no. 100198.
- [41] H. Lee, S. H. Jeong, and H. K. Kim, "OTIDS: A novel intrusion detection system for in-vehicle network by using remote frame," in *Proc. 15th Annu. Conf. Privacy Secur. Trust*, 2017, pp. 57–5709.
- [42] E. C. P. Neto et al., "CICIoV2024: Advancing realistic IDS approaches against dos and spoofing attack in IoV CAN bus," *Internet Things*, vol. 26, 2024, Art. no. 101209.
- [43] O. Y. Al-Jarrah, C. Maple, M. Dianati, D. Oxtoby, and A. Mouzakitis, "Intrusion detection systems for intra-vehicle networks: A review," *IEEE Access*, vol. 7, pp. 21266–21289, 2019.
- [44] A. Rosay, F. Carlier, and P. Leroux, "Feed-forward neural network for network intrusion detection," in *Proc. IEEE 91st Veh. Technol. Conf.*, 2020, pp. 1–6.
- [45] S. Almutlaq, A. Derhab, M. M. Hassan, and K. Kaur, "Two-stage intrusion detection system in intelligent transportation systems using rule extraction methods from deep neural networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 15687–15701, Dec. 2023.
- [46] F. Jin, M. Chen, W. Zhang, Y. Yuan, and S. Wang, "Intrusion detection on Internet of Vehicles via combining log-ratio oversampling, outlier detection and metric learning," *Inf. Sci.*, vol. 579, pp. 814–831, 2021.
- [47] X. Li et al., "Sustainable ensemble learning driving intrusion detection model," *IEEE Trans. Dependable Secure Comput.*, vol. 18, no. 4, pp. 1591–1604, Jul./Aug. 2021.
- [48] Y. Javed, M. A. Khayat, A. A. Elghariani, and A. Ghafoor, "PRISM: A hierarchical intrusion detection architecture for large-scale cyber networks," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 6, pp. 5070–5086, Nov./Dec. 2023.
- [49] M. Abdel-Basset, N. Moustafa, H. Hawash, I. Razzak, K. M. Sallam, and O. M. Elkomy, "Federated intrusion detection in blockchain-based smart transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 3, pp. 2523–2537, Mar. 2022.
- [50] A. Oseni et al., "An explainable deep learning framework for resilient intrusion detection in IoT-enabled transportation networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 1000–1014, Jan. 2023.
- [51] A. Shiravi, H. Shiravi, M. Tavallaei, and A. A. Ghorbani, "Toward developing a systematic approach to generate benchmark datasets for intrusion detection," *Comput. Secur.*, vol. 31, no. 3, pp. 357–374, 2012.
- [52] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. 2015 Mil. Commun. Inf. Syst. Conf.*, 2015, pp. 1–6.
- [53] H. Lu et al., "A transfer learning-based intrusion detection system for zero-day attack in communication-based train control system," *Cluster Comput.*, vol. 27, no. 6, pp. 8477–8492, 2024.
- [54] N. Moustafa, "A new distributed architecture for evaluating AI-based security systems at the edge: Network ton_IoT datasets," *Sustain. Cities Soc.*, vol. 72, 2021, Art. no. 102994.
- [55] H. Mohammadian, A. H. Lashkari, and A. A. Ghorbani, "Poisoning and evasion: Deep learning-based NIDS under adversarial attacks," in *Proc. 21st Annu. Int. Conf. Privacy Secur. Trust*, 2024, pp. 1–9.
- [56] M. L. Han, B. I. Kwak, and H. K. Kim, "Anomaly intrusion detection method for vehicular networks based on survival analysis," *Veh. Commun.*, vol. 14, pp. 52–63, 2018.
- [57] S. Tariq, S. Lee, H. K. Kim, and S. S. Woo, "CAN-ADF: The controller area network attack detection framework," *Comput. Secur.*, vol. 94, 2020, Art. no. 101857.
- [58] Y. Chae, N. Katenka, and L. DiPippo, "An adaptive threshold method for anomaly-based intrusion detection systems," in *Proc. IEEE 18th Int. Symp. Netw. Comput. Appl.*, 2019, pp. 1–4.
- [59] A. Yulianto, P. Sukarno, and N. A. Suwastika, "Improving AdaBoost-based intrusion detection system (IDS) performance on CIC IDS 2017 dataset," *J. Physics: Conf. Ser.*, vol. 1192, 2019, Art. no. 012018.
- [60] M. H. L. Louk and B. A. Tama, "Dual-IDS: A bagging-based gradient boosting decision tree model for network anomaly intrusion detection system," *Expert Syst. Appl.*, vol. 213, 2023, Art. no. 119030.
- [61] M. D. Hossain, H. Inoue, H. Ochiai, D. Fall, and Y. Kadobayashi, "An effective in-vehicle CAN bus intrusion detection system using CNN deep learning approach," in *Proc. IEEE Glob. Commun. Conf.*, 2020, pp. 1–6.
- [62] F. Laghrissi, S. Douzi, K. Douzi, and B. Hssina, "Intrusion detection systems using long short-term memory (LSTM)," *J. Big Data*, vol. 8, no. 1, 2021, Art. no. 65.
- [63] M. K. Putchala, "Deep learning approach for intrusion detection system (IDS) in the Internet of Things (IoT) network using gated recurrent neural networks (GRU)," Master's thesis, Dept. Comput. Sci. Eng., Wright State Univ., Dayton, Ohio, 2017.
- [64] X. Wang et al., "Heterogeneous graph attention network," in *Proc. World Wide Web Conf.*, 2019, pp. 2022–2032.
- [65] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Representations*, 2018, pp. 1–12.
- [66] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Representations*, 2017, pp. 1–14.
- [67] K. Zheng, Q. Zheng, P. Chatzimisios, W. Xiang, and Y. Zhou, "Heterogeneous vehicular networking: A survey on architecture, challenges, and solutions," *IEEE Commun. Surveys Tut.*, vol. 17, no. 4, pp. 2377–2396, Fourth Quarter 2015.
- [68] H. Taslimasa, S. Dadkhah, E. C. P. Neto, P. Xiong, S. Ray, and A. A. Ghorbani, "Security issues in Internet of Vehicles (IoV): A comprehensive survey," *Internet Things*, vol. 22, 2023, Art. no. 100809.
- [69] V. W. Lee et al., "Debunking the 100x GPU vs. CPU myth: An evaluation of throughput computing on CPU and GPU," in *Proc. 37th Annu. Int. Symp. Comput. Archit.*, 2010, pp. 451–460.
- [70] S. A. Tailor, J. Fernandez-Marques, and N. D. Lane, "Degree-quant: Quantization-aware training for graph neural networks," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 1–22.
- [71] H. Zhou, A. Srivastava, H. Zeng, R. Kannan, and V. Prasanna, "Accelerating large scale real-time GNN inference using channel pruning," *Proc. VLDB Endowment*, vol. 14, no. 9, pp. 1597–1605, May 2021.